Module 5: Communication, homeostasis and energy

It is important that organisms, both plants and animals are able to respond to stimuli. This is achieved by communication within the body, which may be chemical and/or electrical. Both systems are covered in detail in this module. Communication is also fundamental to homeostasis with control of temperature, blood sugar and blood water potential being studied as examples.

In this module, the biochemical pathways of photosynthesis and respiration are considered, with an emphasis on the formation and use of ATP as the source of energy for biochemical processes and synthesis of biological molecules.

Learners are expected to apply knowledge, understanding and other skills developed in this module to new situations and/or to solve related problems.

5.1 Communication and homeostasis

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Organisms use both chemical and electrical systems to monitor and respond to any deviation from the body's steady state.

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	the need for communication systems in multicellular organisms	To include the need for animals and plants to respond to changes in the internal and external environment and to coordinate the activities of different organs.
(b)	the communication between cells by cell signalling	To include signalling between adjacent cells and signalling between distant cells.
(c)	the principles of homeostasis	To include the differences between receptors and effectors, and the differences between negative feedback and positive feedback. HSW8
(d)	the physiological and behavioural responses involved in temperature control in ectotherms and endotherms.	To include: • endotherms – peripheral temperature receptors, the role of the hypothalamus and effectors in skin and muscles; behavioural responses • ectotherms – behavioural responses. **PAG11** HSW2